

MĪZAN — Security Policy and Procedures

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This document describes the security practices that govern MĪZAN (<https://mizan-puce.vercel.app>), a personal finance dashboard that aggregates bank, brokerage, and market data via Plaid, SnapTrade, and related read-only APIs.

1. Scope

This policy covers:

- All systems that store, process, or transmit consumer financial data obtained via Plaid or SnapTrade.
- Administrative access to production infrastructure (Vercel, Supabase, GitHub, Plaid Dashboard, SnapTrade Dashboard).
- All code, configuration, secrets, and operational practices used to run the deployed application.

2. Governance

The operator is responsible for information security, including:

- Policy ownership and annual review.
- Vendor risk assessment for any new third-party data processor.
- Incident response and breach notification.
- Maintaining the access control matrix below.

Policy is reviewed annually or whenever a material change in infrastructure, vendors, or data flows occurs (whichever is sooner).

3. Identity and Access Management

3.1 Production system access

- All administrative accounts (Vercel, Supabase, GitHub, Plaid, SnapTrade) require multi-factor authentication.

- Production secrets (Plaid client ID and secret, Supabase service-role key, SnapTrade consumer key, third-party API keys) are stored in Vercel environment variables, scoped per environment (Development, Preview, Production), and never committed to source control.
- Principle of least privilege: the public-facing application uses the Supabase anonymous key with Row-Level Security policies. The service-role key (which bypasses RLS) is used only on the server side and never transmitted to the browser.

3.2 Consumer access

- End users authenticate via Supabase Auth (email + password).
- Multi-factor authentication (TOTP) is available to end users through the in-app account settings.
- Sessions use Supabase-issued JWTs; the server validates the JWT on every request that touches consumer financial data.
- Row-Level Security (RLS) policies enforce that each authenticated user can read only their own rows in Supabase tables.

4. Infrastructure and Network Security

- **Encryption in transit:** TLS 1.2 or better is enforced for all client-to-server traffic via Vercel. HSTS is set in response headers.
- **Encryption at rest:** Supabase Postgres uses AES-256 disk-level encryption. All Plaid access tokens and account metadata are stored in encrypted form.
- **Secret storage:** All API credentials are managed via Vercel encrypted environment variables. No secrets exist in the source repository.
- **Content Security Policy (CSP):** Strict CSP is set via `vercel.json`, including explicit allowlists for Plaid and SnapTrade origins.
- **Security headers:** `Strict-Transport-Security`, `X-Content-Type-Options`, `X-Frame-Options`, `Referrer-Policy`, and `Permissions-Policy` are set on all responses.

5. Development and Vulnerability Management

- **Dependency scanning:** GitHub's automated dependency security alerts (Dependabot) are enabled. Vulnerable dependencies are reviewed and patched on receipt.
- **Patch management:** Upstream package updates are applied through standard `npm` workflows. Production deploys go through Vercel CI.

- **Runtime monitoring:** Sentry is wired in (`lib/sentry.mjs` , `src/main.jsx`) to capture and alert on backend exceptions and frontend errors in production.
- **Anomaly detection:** Custom anomaly detectors (`lib/anomaly.mjs`) monitor authentication failure spikes, SnapTrade error rate spikes, cron staleness, and new-device sign-ins, with email alerts via Resend.
- **Audit logging:** Every security-relevant event (login, MFA enrollment, bank connect, bank disconnect, session revoke, account deletion) is recorded to a Postgres `audit_log` table.

6. Privacy and Data Handling

Refer to the Privacy Policy at <https://mizan-puce.vercel.app/privacy> for the full data-handling disclosure to end users.

6.1 Data categories

- **Authentication data:** email, hashed password (via Supabase Auth), optional TOTP factor.
- **Plaid access tokens:** stored server-side only in `plaid_tokens` (RLS-protected, never returned to the browser).
- **Plaid account metadata:** institution name, account name, masked number, current and available balances, currency. Stored in `plaid_accounts` with RLS.
- **Plaid transactions:** not persisted. Fetched live per request via `/transactions/sync` .
- **SnapTrade user mapping:** opaque user-id and user-secret pair per Supabase user, stored in `user_snaptrade` .

6.2 Retention and deletion

- Account metadata and tokens persist until (a) the user disconnects the institution, or (b) the user deletes their MIZAN account.
- Disconnection calls Plaid `/item/remove` to revoke the access token, then deletes the corresponding rows from `plaid_tokens` and `plaid_accounts` .
- Account deletion: `ON DELETE CASCADE` on `user_id` removes all associated Plaid/SnapTrade rows.
- Retention review is performed at least annually as part of policy review.

6.3 Consent

- End users consent to data collection at signup (acceptance of the Terms of Service and Privacy Policy).

- Plaid Link's own consent flow obtains explicit consent for each institution linked, with the user retaining the ability to disconnect at any time.

7. Incident Response

If the operator becomes aware of a confirmed or suspected security incident affecting consumer financial data:

1. Contain the incident (revoke tokens, rotate credentials, take affected systems offline).
2. Investigate scope and root cause using Sentry, audit logs, and Vercel/Supabase access logs.
3. Notify affected users without undue delay and in accordance with applicable law.
4. Notify Plaid and SnapTrade where their data or systems are implicated.
5. Document the incident, remediation, and lessons learned.

8. Vendor Risk

The following third parties process or store consumer data on behalf of MĪZAN. Each is reviewed for SOC 2 / ISO 27001 / equivalent attestation and a published security policy before integration:

- **Vercel** — hosting, edge network, function execution.
- **Supabase** — Postgres database, authentication, RLS enforcement.
- **Plaid** — bank aggregation.
- **SnapTrade** — brokerage aggregation.
- **Sentry** — error monitoring (PII scrubbing enabled).
- **Resend** — transactional email for security alerts.

9. Contact

Security questions or vulnerability reports: noreply@omni-flow.net

For end-user data requests (access, deletion, correction), refer to the contact channel listed in the Privacy Policy at <https://mizan-puce.vercel.app/privacy>.